

Hardware Resource Usage Logic in Engineering Software

- Hardware Resource: CPU / GPU / RAM / Storage / Network
- This document summarizes the complete hardware resource usage pattern of engineering software—from startup to rendering/solving—and explains the resource dependencies of different software categories.

1. Software Startup (Loading Program)

- **Primary load: Storage (SSD) → Memory (RAM)**
- Load executable files, DLLs, plugins
- Initialize UI and caches
- CPU/GPU barely involved
- **Bottleneck: Slow SSD → slow startup**

2. Opening Models / Assemblies (Loading Model)

- **Primary load: Memory (RAM) → Storage (SSD)**
- Load geometry, materials, textures
- Load BIM parameter trees, CAD assembly trees
- Large models consume large RAM
- **Bottleneck: Insufficient RAM → excessive disk swapping → stutter**

3. Modeling / Editing (Sketch / Boolean / Feature)

- **Primary load: CPU single-core performance (most critical)**
- Sketch constraint solver
- Boolean operations
- Feature regeneration
- Parametric updates
- **Bottleneck: Low CPU single-core frequency → lag**

4. Viewport Display / Model Rotation (Viewport)

- **Primary load: GPU (professional graphics card)**
- OpenGL/DirectX rendering
- Shadows, transparency, anti-aliasing
- Large model rotation
- **Bottleneck: Weak GPU → low FPS**

5. Large Models / Large Assemblies (Large Model / Assembly)

- **Primary load: Memory (RAM)**
- Large BIM models (hospitals, airports)
- Large CAD assemblies (cars, aircraft)
- DMU (Digital Mock-Up)
- **Bottleneck: Insufficient RAM → crashes**

6. Solving (CAE Solver)

- **Primary load: Multi-core CPU + RAM + SSD write speed**
- Mesh generation
- Solver iterations
- Writing large result files (tens of GB)
- **Bottleneck: Insufficient RAM → solver failure**

7. Rendering (Render)

- **CPU rendering: uses multi-core CPU**
- **GPU rendering: uses GPU VRAM + CUDA cores**
- Ray tracing
- Path tracing
- Material evaluation
- **Bottleneck: Insufficient VRAM → render failure**

8. Collaboration / PLM (ENOVIA / Teamcenter)

- **Primary load: Network + server performance**
- File synchronization
- BOM
- Permissions
- Version control
- **Bottleneck: High network latency → slow operations**

Core Hardware Usage Rules (Top 8 Universal Patterns)

1. Startup → SSD
2. Model loading → RAM
3. Modeling → CPU single-core
4. Display → GPU
5. Large models → RAM
6. Solving → Multi-core CPU + RAM
7. Rendering → GPU (or multi-core CPU)
8. Collaboration → Network

Hardware Dependency Differences Across Software Categories

BIM (Revit / Archicad / Tekla)

- CPU: single-core for modeling and quantity takeoff
- GPU: moderate GPU for viewport
- RAM: most critical (large models consume huge RAM)
- SSD: model loading
- Network: collaboration workflows

CAD (CATIA / SolidWorks / NX / AutoCAD)

- CPU: heavy single-core dependency
- GPU: professional GPU for viewport

- RAM: large assemblies require large RAM
- SSD: affects assembly loading speed
- Network: minimal dependency

CAM (Mastercam / UG CAM / CATIA Machining)

- CPU: toolpath calculation (single-core or light multi-core)
- GPU: viewport display
- RAM: moderate requirement
- SSD: toolpath caching

CAE (ANSYS / Abaqus / Nastran)

- CPU: strong multi-core dependency
- RAM: extremely high requirement (million-element models)
- SSD: solver output files
- GPU: some solvers support CUDA

Rendering (Blender / KeyShot / 3ds Max)

- GPU: most critical (VRAM + CUDA)
- CPU: multi-core for CPU rendering
- RAM: textures and scene caching
- SSD: cache writing

Real-time Engines (Unreal / Unity)

- GPU: real-time rendering
- CPU: scene logic
- RAM: scene resources
- SSD: loading speed

What Mainstream Software Can Do at Different Hardware Levels

BIM (Revit / Archicad / Tekla)

Low-end (can run)

- **CPU:** i5-13400 / i5-12400
- **GPU:** RTX 3050 / GTX 1660 Super
- **RAM:** 16GB
- Small residential models
- Basic modeling
- Simple families
- Small project documentation

Mid-range (mainstream engineers)

- **CPU:** i7-13700K / i7-14700K
- **GPU:** RTX A2000 / RTX 3060 / RTX 4060 Ti

- **RAM:** 32GB
- Medium commercial buildings
- Multi-discipline collaboration
- Heavy family usage
- Complex views

High-end (large BIM projects)

- **CPU:** i9-13900K / i9-14900K
- **GPU:** RTX A4000 / RTX A4500 / RTX 4070 Ti
- **RAM:** 64GB
- Large hospitals, airports
- 1GB+ BIM models
- Heavy schedules
- Complex render views

CAD (CATIA / SolidWorks / NX / AutoCAD)

Low-end

- **CPU:** i5-13400 / i5-12600K
- **GPU:** T1000 / RTX 3050
- **RAM:** 16GB
- Small part modeling
- Simple assemblies

Mid-range

- **CPU:** i7-13700K / i7-14700K
- **GPU:** RTX A2000 / RTX A4000 / RTX 4060 Ti
- **RAM:** 32GB
- Medium assemblies
- Surface modeling
- Heavy drawings

High-end

- **CPU:** i9-13900K / i9-14900K
- **GPU:** RTX A5000 / RTX A4500 / RTX 4080
- **RAM:** 64GB
- Large assemblies
- DMU
- Advanced surfacing

CAM (Mastercam / UG CAM / CATIA Machining)

Low-end

- **CPU:** i5-13400
- **GPU:** RTX 3050

- **RAM:** 16GB
- 2.5-axis machining
- Simple toolpaths

Mid-range

- **CPU:** i7-13700K
- **GPU:** RTX A2000 / RTX 3060
- **RAM:** 32GB
- 3-axis machining
- Medium parts

High-end

- **CPU:** i9-13900K
- **GPU:** RTX A4000 / RTX 4070 Ti
- **RAM:** 64GB
- 5-axis machining
- Large molds
- Complex toolpaths

CAE (ANSYS / Abaqus / Nastran)

Low-end

- **CPU:** i7-12700 / Ryzen 7 5800X
- **GPU:** optional
- **RAM:** 32GB
- Small models
- Linear analysis

Mid-range

- **CPU:** Ryzen 9 7950X / i9-13900K
- **GPU:** RTX 3060
- **RAM:** 64GB
- Medium models
- Thermal / modal analysis

High-end

- **CPU:** Threadripper Pro 5975WX / Threadripper Pro 5995WX
- **GPU:** RTX A5000
- **RAM:** 128GB–256GB
- Million-element models
- Nonlinear analysis
- Explicit dynamics

Rendering (Blender / KeyShot / 3ds Max)

Low-end

- **CPU:** i5-13400
- **GPU:** RTX 3050
- **RAM:** 16GB
- Small scenes
- Low-res rendering

Mid-range

- **CPU:** i7-13700K
- **GPU:** RTX 3060 / RTX 3070 / RTX 4060 Ti
- **RAM:** 32GB
- Medium scenes
- GPU rendering

High-end

- **CPU:** i9-13900K / Ryzen 9 7950X
- **GPU:** RTX 4090 / RTX A6000
- **RAM:** 64GB–128GB
- Large scenes
- 8K rendering
- Animation rendering

Real-time Engines (Unreal / Unity)

Low-end

- **CPU:** i5-12400
- **GPU:** RTX 3050
- **RAM:** 16GB
- Small scenes
- Low settings

Mid-range

- **CPU:** i7-13700K
- **GPU:** RTX 3070 / RTX 4060 Ti
- **RAM:** 32GB
- Medium scenes
- High settings

High-end

- **CPU:** i9-13900K
- **GPU:** RTX 4080 / RTX 4090
- **RAM:** 64GB
- Large scenes
- Real-time ray tracing
- Industrial VR/AR